

HARIKA MANUKONDA

AI/ML Engineer | Data Scientist | MLOps Specialist

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SUMMARY

AI/ML Engineer and Data Scientist with **3+ years** of experience delivering scalable **machine learning, MLOps, Generative AI applications, and LLMops** solutions across McKesson, Airbnb, and Dell Technologies. Skilled in building and governing production-grade AI systems with **robust model governance and compliance frameworks**. Skilled in **Python, TensorFlow, PyTorch**, and **LangChain**, with hands-on expertise in **AWS SageMaker, Azure ML**, and **Snowflake**. Proven ability to build **end-to-end ML pipelines**, optimize **model performance**, and deploy **production-grade AI systems** that drive automation and data-driven decision-making. Known for strong communication, analytical thinking, and cross-functional collaboration skills.

PROFESSIONAL EXPERIENCE

AI/ML Engineer / Data Scientist | McKesson, USA | Aug 2025 – Present

- **Developed** multimodal forecasting systems using **Python, TensorFlow, and OpenAI API**, training hybrid **LLMs and predictive models** on **10M+ logistics transactions** to cut supply-chain disruption incidents by **2,300 annually** across critical healthcare operations.
- **Constructed** advanced **Retrieval-Augmented Generation (RAG)** pipelines with **LangChain, Hugging Face, LlamaIndex, and FAISS**, integrating domain-tuned **medical text embeddings** that automated clinical report summarization across **5 enterprise systems**, saving **7,800 labor hours yearly**.
- **Orchestrated** production-grade **MLOps and GenAIOps workflows** using **AWS SageMaker, MLflow, Docker, and Kubernetes**, embedding **human feedback loops** and **model governance tracking**, ensuring full compliance across **12 deployed AI services** within McKesson's regulated data ecosystem.
- Integrated **LLMOps workflows** for managing model lifecycle, compliance, and observability across Generative AI deployments.

AI/ML Engineer | Airbnb, India | Dec 2022 – Aug 2023

- **Developed large-scale ML data pipelines** using **Python (Pandas, PySpark)** and **SQL**, automating ingestion from **Snowflake and AWS S3**, and processing **5.3M+ daily transactions** for forecasting and anomaly detection models.
- **Trained and optimized machine learning models** in **TensorFlow, scikit-learn, and XGBoost**, achieving measurable gains in **demand prediction accuracy** and improving operational planning for regional market analytics.
- **Administered automated data preprocessing and feature engineering** pipelines with **dbt and Airflow**, improving **model retraining speed** and ensuring schema consistency across evolving datasets.
- **Deployed ML inference workflows** using **Docker, FastAPI, and AWS Lambda**, enabling **real-time model serving** and reducing prediction latency by **35%** across live traffic streams.
- **Monitored model performance and data drift** through **MLflow and Prometheus**, maintaining version control, experiment tracking, and reproducibility across all production environments.
- **Collaborated with cross-functional teams** to build **3 ML-ready data marts** and deploy **2 production AI models**, aligning analytical outcomes with business KPIs and accelerating **AI adoption across Airbnb's data ecosystem**.

ML Engineer | Dell Technologies, India | Jun 2020 – Nov 2022

- **Built and automated end-to-end ML pipelines** using **Azure Data Factory, Python, and SQL**, processing **1.5M+ daily transactions** to support model training, validation, and continuous performance monitoring.
- **Implemented robust MLOps practices** with **Azure ML, Docker, and GitHub Actions**, automating **model retraining and deployment** across **4 production clusters**, reducing delivery time by **70 hours per month**.
- **Engineered advanced feature extraction and transformation workflows** using **PySpark, Pandas, and NumPy**, improving forecasting model precision across **10 global business segments**.
- **Crafted and tuned predictive algorithms** (ARIMA, XGBoost, Random Forest) for **demand forecasting and anomaly detection**, enhancing model accuracy and business scheduling efficiency.
- **Integrated cloud and on-prem data systems** through **Azure Synapse, SQL Server, and REST APIs**, ensuring **secure model access, experiment reproducibility**, and low-latency analytics for ML workloads.
- **Deployed model performance monitoring and drift detection pipelines** using **MLflow, Prometheus, and Azure Monitor**, enforcing **AI governance standards** and maintaining version-controlled model transparency.

TECHNICAL SKILLS

- **Programming & Analytics:** Python (Pandas, NumPy, SciPy), R, SQL, MATLAB, Excel (Power Query, Pivot Tables), Statistical Modeling, Hypothesis Testing, Time Series Forecasting
- **Machine Learning & AI:** Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, Decision Trees, Random Forest, SVM, Ensemble Methods, Feature Engineering, Model Evaluation

- **Generative AI & LLMs:** OpenAI GPT Models (GPT-4, GPT-3.5), LangChain, Hugging Face Transformers, Prompt Engineering, RAG (Retrieval-Augmented Generation), LlamaIndex, Vector Databases (Pinecone, FAISS, ChromaDB)
- **Data Visualization & BI Tools:** Power BI, Tableau, Matplotlib, Seaborn, Plotly, Looker Studio, Excel Dashboards, DAX, KPI Reporting, Data Storytelling
- **Data Engineering & Pipelines:** Apache Airflow, Alteryx, ETL Development, PySpark, Azure Data Factory, AWS Glue, Snowflake, Databricks, Data Cleaning & Transformation
- **Big Data & Cloud Ecosystems:** AWS (S3, Redshift, SageMaker, Lambda), Azure (Synapse, Machine Learning, Data Lake), Google Cloud (BigQuery, Vertex AI), Docker, Kubernetes
- **Databases & Querying:** MySQL, PostgreSQL, SQL Server, MongoDB, Snowflake, Cassandra, Data Warehousing
- **NLP & Deep Learning:** Transformers, BERT, T5, LSTM, Seq2Seq, Text Classification, Named Entity Recognition (NER), Sentiment Analysis, Embeddings, Tokenization
- **MLOps, LLMops & Model Governance:** MLflow, Weights & Biases, FastAPI, Model Monitoring, API Deployment, CI/CD Integration (GitHub Actions, Jenkins, Docker, Kubernetes)
- **Data Governance & Automation:** Data Quality, Data Cataloging, Power Automate, Validation Scripts, Process Automation, Version Control (Git, GitHub, GitLab)
- **Soft Skills:** Leadership, Problem-Solving, Collaboration, Adaptability, Communication

PROJECTS

Healthcare Supply-Chain AI Optimization

- **Engineered** multimodal forecasting and patient-risk models using **TensorFlow, Hugging Face, and OpenAI API**, processing **10M+ logistics and clinical records** to prevent **2,300 supply-chain disruptions** and identify **early-stage cancer cases**.
- **Built** scalable **RAG pipelines and predictive analytics workflows** in **AWS Sage Maker, Snowflake, and MLflow**, automating **medical text summarization across 5 systems** and eliminating **15,800 manual tasks annually** through strategic **collaboration** with clinical data teams.

Shop for Home

- **Developed** a scalable full-stack **e-commerce platform** using **React.js, Node.js, and MongoDB**, enabling real-time order management for **3,000+ SKUs** and supporting **thousands of concurrent users** with seamless authentication and secure access control.
- **Architected** **microservices-based order and inventory synchronization** with **Express.js and REST APIs**, ensuring **zero data conflicts** and achieving **sub-second response times** for critical transactions, demonstrating strong **problem-solving** and technical design skills.

Monthly Salary Prediction

- **Designed** and trained a **salary prediction model** using **Linear Regression and Random Forest**, analyzing **250,000+ industry records** to achieve **88% prediction accuracy** for workforce compensation forecasting.
- **Deployed** automated model inference with **Flask APIs** and integrated **salary trend visualization dashboards**, cutting **manual HR analytics time by 120 hours monthly** through intuitive **data storytelling** and performance reporting.

CERTIFICATIONS

- **AWS Certified AI Engineer – Practitioner** (*Expected: Dec 2025*)
- **AWS Certified Machine Learning – Specialty** (*Expected: Dec 2025*)
- **Databricks Certified Generative AI** (*Expected: Jan 2026*)

EDUCATION

Saint Louis University

Master's in information systems | GPA: 3.8 / 4.0

St. Louis, MO

Aug 2023 – May 2025